

Demonstration and Viva Guide

Haunted Toy Room: The Midnight Mission

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Start and rehearse

Run `bin/Release/HauntedToyRoom.exe` from the project folder. The arrival starts automatically. Use Y to reach the toy room quickly, Shift+N for a full replay, and G to hide the interface. Press H for help. Keep the report PDF, slide deck and Project-Demo.mp4 together for the showcase.

Two-minute video narration

The video has visual captions and no recorded voice. Explain the following points live. The deck embeds the video on slide 2; the standalone MP4 works independently of slide playback support.

Time	Explain
0:00–0:10	Welcome home: Penny's route links the garden, doorway, stairs and toy room.
0:10–0:20	A furnished world: Five reusable primitives build the house, furniture and articulated toys.
0:20–0:32	Hierarchy in motion: Jessie attaches to Bullseye's saddle; one parent transform moves both.
0:32–0:44	Flight, laser and collision: Buzz aims at the obstacle. The nearest ray hit receives an impulse.
0:44–0:56	Rescue and return: The car follows waypoints. Wheel rotation uses distance divided by radius.
0:56–1:06	Live character takeover: Woody follows your controls while Jessie and Buzz continue the mission; O releases him, N enters full manual.
1:06–1:14	Directional, point and spot: Moonlight is directional. The lamp adds a point source and a soft cone.
1:14–1:20	Gouraud shading: Light is evaluated at vertices; the rasterizer interpolates the result.
1:20–1:26	Phong shading: Interpolated normals are normalized; the reflection-vector highlight is per pixel.
1:26–1:34	Procedural textures: UVs map grain, wallpaper, fabric and lunar craters onto shared geometry.
1:34–1:46	Ray tracing: A BVH finds analytic hits; shadow rays and mirror bounces reveal the polished floor.
1:46–1:52	A quiet morning: The lamp turns off, the toys return home and Penny falls asleep.
1:52–2:00	Thank you: Adiba Tahsin Roll 2107031 CSE-4102

Live demonstration sequence

1. Y skips arrival; 1 selects Woody for live takeover. Hold W, turn with A/D, stop with Space. Jessie and Buzz continue their routes. Press 0 to release Woody; press N for full manual mode. Explain separate input ownership and scene-clock progression.
2. 2 selects Jessie. R is distance-gated at 2.2 units. Move her near Bullseye or use the reproducible mounted launch below, then R. Select either rider or horse and drive; R dismounts.
3. 4 selects Buzz. Q/E moves vertically, L toggles the laser, Z/X aims, B rebuilds blocks. Explain the emitter hierarchy, nearest-hit test and impulse.
4. 5 selects the car. W/S and A/D drive, L toggles moving spotlights. F focuses it; hallway camera access is supported.
5. 7 selects the lamp. R toggles power, W/S tilts, A/D swivels, comma/period alters intensity. Both light position and direction follow the rig.
6. F2 cycles Flat/Gouraud/Phong/Blinn and selects the raster path. F5/F6/F7 isolate terms. F3 toggles colour textures. F4 selects analytic ray tracing; 9 changes bounces and minus/equal changes resolution.
7. Click a toy or furniture. Tab enters edit mode; T chooses Translate/Rotate/Scale/Shear. J/L, U/O and I/K change axes. M mirrors, Backspace restores. V lists actual parts; Shift+V prints vertices/indices.
8. O enables haunted ambience. Show the ghost's opacity, ball rolling and lamp motion. N resumes the coordinated story; Shift+N restarts it.

Reproducible rider setup:

```
.bin\Release\HauntedToyRoom.exe --no-intro --manual --mount --select 1 --no-raytrace
```

Reproducible matched shading view:

```
.bin\Release\HauntedToyRoom.exe --no-intro --no-raytrace --shading 1 --select 3 --cam  
0.4,1.5,0.5,0.4,1.15,-1.8
```

Live control design

Selection during playback owns only one actor. StoryDirector skips writes to that actor and advances a separate virtual route cursor for scene gates. Other actors continue normal waypoint motion and animation. They ignore the owned actor as an obstacle, so parking across a path cannot stall their choreography; wall and furniture contacts remain active for your actor. Buzz takeover uses a three-second doorway gate without changing your laser or aim. A mounted pair is detached when selected for independent control; remounting intentionally drives the connected pair. 0 releases ownership; N retains the separate full manual mode. Penny is controllable after arrival by clicking her. The arrival camera remains cinematic until Y or arrival completion.

Questions you should answer

Why 24 cube vertices instead of eight?

A geometric corner belongs to three faces. Each face needs a distinct normal and UV chart, so the indexed mesh keeps four vertices per face.

What is the difference between illumination and shading?

Illumination computes light at one point. Shading decides where that computation is evaluated and which quantities are interpolated.

Why can Gouraud miss a highlight?

It evaluates specular response only at vertices. A peak between vertices is absent from the interpolated values.

Why normalise a Phong normal after interpolation?

Interpolation does not preserve unit length. The cosine dot product requires a unit vector.

Why inverse transpose for normals?

It preserves perpendicularity to transformed tangents under non-uniform scale and shear: $(A^T \cdot n) \cdot (A \cdot t) = n \cdot t$.

What happens when shininess increases?

The exponent suppresses values below one more strongly, producing a narrower specular highlight. k_s controls its strength.

Why are the spotlight cutoffs cosine values?

The cone test uses a dot product of unit vectors. The outer angle is larger, so its cosine is smaller; smoothstep fades between these bounds.

How does Jessie follow Bullseye?

Her root is attached under the saddle. The horse world matrix multiplies the saddle and rider local matrices, so translation, rotation and gait propagate naturally.

Why do you not normalise the transformed ray direction?

Keeping its length preserves the same t in object/world space. Hits from different scaled instances can then be compared directly.

Is the tracer testing every triangle?

No. The BVH bounds scene instances, then tests exact plane, cube, sphere, cylinder and cone equations in object space.

Is transparency the same as refraction?

No. This implementation continues straight through with reduced throughput. It does not apply Snell's law.

How are raster shadows different from ray shadows?

Raster lamp shadows compare a receiver with the lamp depth map. Ray shadows query whether an opaque object lies before the selected light.

Why retain off-screen geometry in the ray scene?

It may still appear in a reflection or occlude a light. Camera frustum culling is applied only to the raster submission.

Does a colour texture change geometry?

No. It modulates albedo. The fence alpha is coverage, while the moon craters and cloth weave remain colour detail rather than displacement or normal mapping.

How are wheel and ball angles calculated?

Angle equals distance/radius. A ball uses an axis perpendicular to up and displacement; wheels rotate around their fixed axle.

How do optimisations preserve the appearance?

Shared buffers preserve identical geometry; LOD retains more triangles when large on screen; conservative bounds prevent false culling; texture detail replaces subpixel objects; bounded simulation prevents unbounded catch-up work.

What are the deliberate limitations?

Lamp-only general raster shadow map; selected/thresholded ray shadows; finite continuations; straight transparency; box-approximate contacts. No global diffuse transport or physical refraction.

Parameter-change practice

Parameter	Location	Expected effect
`shininess` / `ks`	`Room.cpp` material creation or `Material.h` default	Narrower highlight / brighter highlight
`linear`, `quadratic`	`ToyRoomApp::BuildScene` light setup	Faster dimming with distance
22° / 34° lamp cone	`ToyRoomApp::BuildScene` cutoff values	Inner lit region / larger smooth cone
`reflectivity`	`Room.cpp` floor/ball materials	Stronger mirror contribution in tracer

Parameter	Location	Expected effect
`rayBounces`, `rayScale`	`RenderSettings.h`; keys 9 and minus/equal	Deeper continuation / more pixels and cost
`maxSpeed`, `turnRate`	`Character.h`	Faster motion / faster turning
Sphere stacks/sectors	`Assets::Load`	Smoother silhouette but more triangles
`uvScale`	Material creation in room/house builders	More/fewer pattern repeats
Fan 110 degrees/s	`ToyRoomApp::StepScene`	Faster/slower rotor animation

One frame: code reading route

Read `Application::Run` → `ToyRoomApp::OnUpdate` / `StepScene` → input handlers → `StoryDirector::Update` and `Character::Animate` → `PhysicsWorld` → `SceneNode::UpdateWorld` → `UpdateLights` → `OnRender` → `Renderer::Render` or `RayTracer::Render`. Read `lighting.glsl` once, then compare `gouraud.vert`, `lit.frag` and `raytrace.frag`. Look at the named object's builder and inventory row when explaining a part. You do not need to memorise every line.

Checks and reproduction

```
.\tools\build.ps1 -Configuration Release
```

```
.\tools\build.ps1 -Configuration Debug
```

```
.\tools\check-physics.ps1
```

```
python tools/check_showcase_interaction.py
```

```
python tools/check_showcase_live.py
```

```
python tools/make_showcase.py --capture --video
```

```
python tools/build_showcase_docs.py
```

Document generation requires pdfLaTeX (MiKTeX or TeX Live) as well as Python packages listed in `tools/showcase-requirements.txt`. The final report source is `Project-Report.tex`; `python tools/build\_latex\_report.py` rebuilds the PDF alone. Existing artifacts are ready to use; regeneration is optional. `validation/` contains build/check, interaction, story, capture and media evidence. `inventory/` is the complete node/material/surface-map data used for the report.